SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO2 ASSESSMENT TOOL 1 – Scenario-Based Requirement Analysis

Course Code:	CSA10 / CSA1007	**Course Title:**	Software Engineering
CO Assessed:	CO2	**Assessment:**	Assessment Tool 1 – Scenario-Based Requirement Analysis
Weightage:	20%	**Total Marks:**	25
CO2 Statement:	*Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills*		
Student Name:	Sk. Parvez Ahamed	**Reg. No.:**	192411084
Date:	28/07/2026	**Duration:**	60 minutes

Code of Conduct

I _________Sk.Parvez Ahamed (192411084)______ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _______Sk. Parvez Ahamed_________

Annexure A – Scenario-Based Requirement Analysis

Instructions to Students:
Each student will be assigned an individual software project problem statement. Analyze the given scenario and prepare a complete Software Requirement Analysis document by following the steps below.

Question

Based on the **assigned problem statement**, perform a **Scenario-Based Requirement Analysis** and prepare a Software Requirement Specification (SRS) document. Your analysis should include the following:

1. **Study and Analyze the Scenario**
   - Carefully understand the given problem statement.
   - Identify the business domain, stakeholders, existing challenges, and system objectives.
2. **Identify Functional and Non-Functional Requirements**
   - List all functional requirements required by the proposed system.
   - Identify suitable non-functional requirements such as performance, security, reliability, usability, scalability, maintainability, and availability.

3. **Identify Actors and Use Cases**
    - Identify all primary and secondary actors interacting with the system.
    - List the major use cases associated with each actor.
    - Draw a Use Case Diagram representing the interactions between actors and the system.
4. **Prepare the Software Requirement Specification (SRS)**
   Develop an SRS document containing:
    - Introduction
    - Problem Description
    - Scope of the System
    - Functional Requirements
    - Non-Functional Requirements
    - Use Case Descriptions
    - Data Requirements
    - External Interface Requirements
    - System Features
5. **Identify Assumptions and Constraints**
    - List the assumptions considered while developing the system.
    - Identify technical, operational, economic, legal, and time-related constraints affecting the project.
6. **Validate Requirement Completeness and Consistency**
    - Verify whether all stakeholder requirements have been addressed.
    - Check for ambiguity, redundancy, inconsistency, and missing requirements.
    - Suggest improvements to enhance the quality and completeness of the requirement specification.

S.No : 13

Blockchain-Based Academic Credential Verification Platform

1. Study and Analyze the Scenario

1.1 Introduction

Academic certificates are essential documents used for employment, higher education, scholarships, and professional certifications. Traditional verification methods are manual, time-consuming, and vulnerable to document forgery. The proposed Blockchain-Based Academic Credential Verification Platform provides a secure digital system where certificates are issued, stored, and verified using blockchain technology. Each certificate is digitally signed and its hash is stored on the blockchain, ensuring authenticity and preventing tampering.

1.2 Business Domain

The proposed system belongs to the Education Technology (EdTech) domain and is applicable to:

- Universities and Colleges

- Government Education Departments
- Recruitment Organizations
- Scholarship Verification
- Professional Certification Agencies

1.3 Existing Challenges

The current system faces several challenges:

- Manual certificate verification
- Fake academic certificates
- High administrative workload
- Slow verification process
- Risk of certificate tampering

1.4 System Objectives

The proposed platform aims to:

- Issue secure digital certificates.
- Prevent certificate forgery.
- Enable instant verification.
- Maintain tamper-proof records.
- Support multiple institutions.
- Generate verification reports.

2. Functional and Non-Functional Requirements

2.1 Functional Requirements

- User Registration and Login
- Student Management
- Certificate Issuance
- Blockchain Storage
- QR Code Generation
- Certificate Verification
- Report Generation
- Audit Trail

2.2 Non-Functional Requirements

- **Security:** Encryption, blockchain, digital signatures.
- **Performance:** Fast certificate verification.
- **Reliability:** High availability.
- **Scalability:** Support multiple institutions.
- **Usability:** User-friendly interface.

3. Identify Actors and Use Cases

3.1 Primary Actors

- Student
- University Administrator
- Employer
- System Administrator

3.2 Secondary Actors

- Government Authority
- Blockchain Network
- Database Server
- Email Notification Service

3.3 Major Use Cases Student

- Register
- Login
- View and Download Certificate
- Share Certificate

University

- Register Student
- Issue Certificate
- Update Certificate
- View Reports

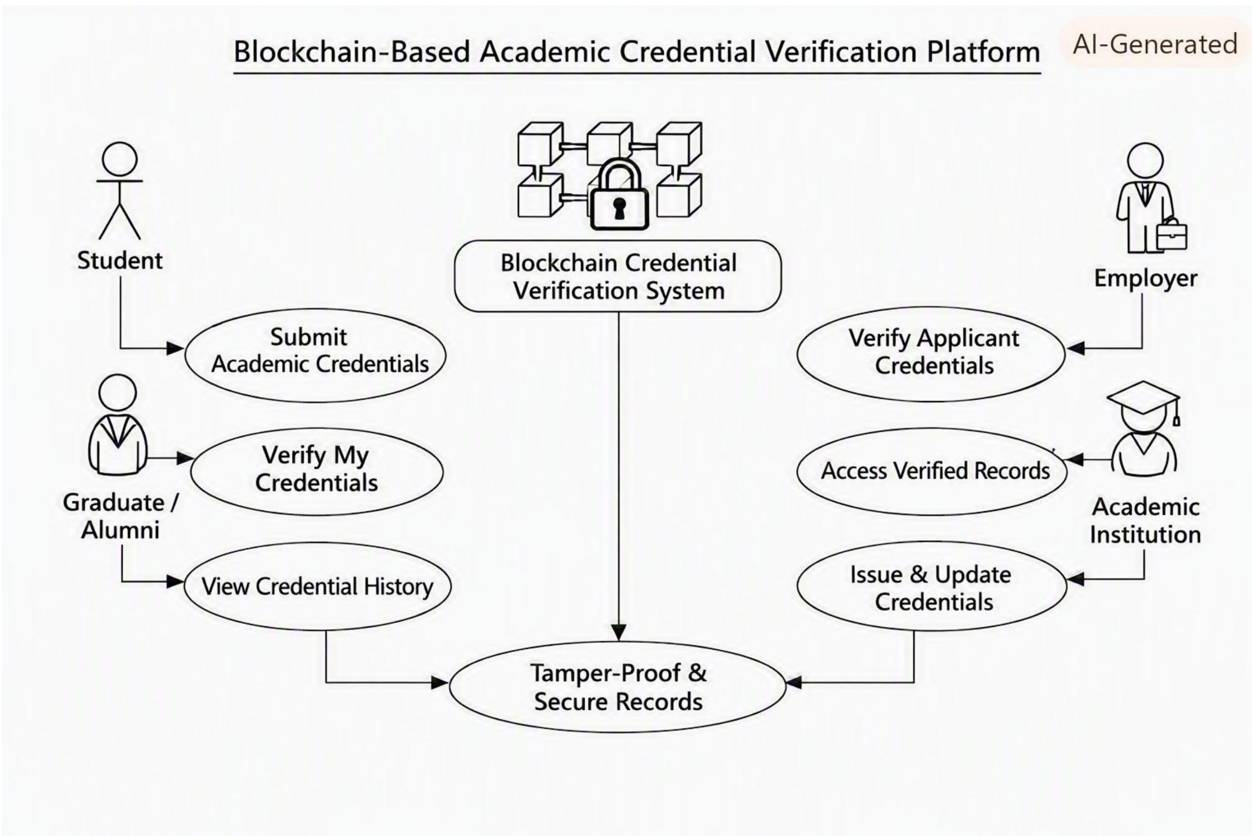
Employer

- Verify Certificate

- Scan QR Code
- Download Verification Report

Administrator

- Manage Users
- Manage Institutions
- Generate Reports
- Monitor Audit Logs



4. Software Requirement Specification (SRS)

4.1 Introduction

The proposed platform improves the security and efficiency of issuing and verifying academic certificates. It replaces manual verification with a blockchain-based system that enables secure and instant verification.

4.2 Problem Description

Traditional certificate verification is slow, manual, and vulnerable to forgery. The proposed platform addresses these issues by storing certificate information securely on the blockchain.

4.3 Scope of the System The platform includes:

- Certificate Issuance
- Blockchain Storage
- Student Portal
- Employer Verification
- QR Code Verification
- Report Generation
- Audit Management

4.4 Functional Requirements The system shall:

- Register and authenticate users.
- Issue digital certificates.
- Store certificate hashes on blockchain.
- Generate QR codes.
- Verify certificates.
- Generate reports.
- Maintain audit logs.

4.5 Non-Functional Requirements The system shall provide:

- High Security
- Good Performance
- Reliability
- Scalability
- Maintainability
- User-Friendly Interface

4.6 Use Case Descriptions

Use Case 1 – Issue Certificate

Primary Actor: University Administrator

Precondition: Student details are verified.

Flow

1. Login
2. Select Student
3. Generate Certificate
4. Store Hash on Blockchain
5. Generate QR Code

Postcondition

Student receives a verified digital certificate.

Use Case 2 – Verify Certificate
Primary Actor: Employer

Precondition: Employer has the Certificate ID or QR Code.

Flow

1. Open Verification Portal
2. Enter Certificate ID
3. Verify using Blockchain
4. Display Result

Postcondition

Certificate authenticity is confirmed.

4.7 Data Requirements Student Data

- Student ID
- Name
- Degree
- Department

Certificate Data

- Certificate ID
- Certificate Hash
- QR Code

Institution Data

- University Name
- Institution ID

Verification Data

- Verification Date
- Status
- Transaction ID

4.8 External Interface Requirements
User Interface

- Web Portal
- Mobile Support

Hardware Interface

- Computer
- Smartphone

Software Interface

- Blockchain Network
- Database

Communication Interface

- HTTPS
- REST APIs

4.9 System Features

- Secure Login
- Certificate Issuance
- Blockchain Verification
- QR Code Verification
- Report Generation
- Audit Trail

5. Assumptions and Constraints

5.1 Assumptions

- Universities maintain accurate records.
- Internet connectivity is available.
- Blockchain network is operational.
- Users follow security guidelines.

5.2 Constraints Technical

- Blockchain integration complexity.

Operational

- User training required.

Economic

- Initial implementation cost.

Legal

- Data privacy regulations.

Time

- Fixed project schedule.

6. Validate Requirement Completeness and Consistency

Requirement Validation

- Completeness: All major stakeholder requirements are included.
- Consistency: No conflicting requirements exist.
- Feasibility: Requirements are technically achievable.
- Traceability: Requirements align with stakeholder needs.

Suggested Improvements

- AI-based fraud detection
- Mobile application
- Biometric authentication
- Government database integration

Conclusion

The Blockchain-Based Academic Credential Verification Platform provides a secure, scalable, and transparent solution for issuing and verifying academic credentials. By leveraging blockchain technology, digital signatures, and QR-code-based verification, the system eliminates certificate fraud, reduces verification time, and improves trust among universities, students, employers, and government authorities. The proposed SRS defines the functional and non-functional requirements, identifies all stakeholders and use cases, specifies system interfaces, and validates requirement completeness, making it a strong foundation for the subsequent design and implementation phases.

Rubrics for Calculation

Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Scenario Understanding and Problem Analysis	Demonstrates thorough understanding of the scenario, stakeholders, objectives, and business context with clear analysis.	Demonstrates good understanding with minor omissions.	Basic understanding with limited analysis.	Poor understanding; major aspects missing or incorrect.
Functional and Non-Functional Requirement Identification	Clearly identifies comprehensive, accurate, and well-classified functional and non-functional requirements.	Identifies most requirements with minor classification errors.	Identifies only basic requirements; several omissions.	Requirements are incomplete, inaccurate, or poorly classified.
Actor Identification and Use Case Modeling	Correctly identifies all relevant actors and use cases; use case diagram is complete, accurate, and follows UML standards.	Most actors and use cases identified; diagram has minor errors.	Limited actors/use cases identified; diagram partially correct.	Actors/use cases missing or incorrect; diagram absent or inaccurate.
Software Requirement Specification (SRS) Documentation	SRS is well-structured, complete, professionally organized, and includes all required sections.	SRS is organized with minor missing sections or formatting issues.	SRS includes only essential sections with limited detail.	SRS is incomplete, poorly organized, or lacks essential components.
Assumptions, Constraints, and Requirement Validation	Clearly identifies realistic assumptions and constraints; validates requirements for completeness, consistency, and ambiguity with appropriate justification.	Identifies most assumptions and constraints; validation is adequate.	Limited assumptions/constraints; validation is superficial.	Assumptions, constraints, and validation are missing or incorrect.

Criteria	**Mark**
Scenario Understanding and Problem Analysis	/5
Functional and Non-Functional Requirement Identification	/5
Actor Identification and Use Case Modeling	/5
Software Requirement Specification (SRS) Documentation	/5
Assumptions, Constraints, and Requirement Validation	/5
TOTAL	/5